

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and developed **back-end web applications** using **Java Spring Boot** and **PHP**, delivering REST APIs and reusable code libraries for laboratory information systems serving 50+ concurrent users in production environments.
- Evolved platform functionality through **continuous improvement** initiatives: automated **data computations**, optimized **data transfer between instruments**, and implemented **data quarantine systems** to handle 30% increase in monthly data throughput while maintaining **automated quality control verifications**.
- Designed and deployed **integration / data import / export systems** using **Python ETL pipelines** and **SQL** (PostgreSQL, MySQL, Oracle) to synchronize data across on-premise and cloud infrastructure; managed **document release deployment & change management** workflows via **Jenkins CI/CD** and **Git**.
- Performed **statistical analysis** for **bias, precision, and uncertainty of measurement controls** using **R / RStudio** and **Python** (pandas, NumPy, scikit-learn); developed **Power BI dashboards** with **DAX measures** and **data modelling** for cross-functional teams in quality and analytics.
- Troubleshoot and resolved production issues across **on premise & cloud** environments using **AWS (EC2, S3, Lambda, CloudWatch)**, **Docker containerisation**, and **Linux** system administration; maintained SLA compliance and documented **design solutions** for new feature development in technical specifications.
- Met with **cross-functional teams** in analytical, quality, and field services to understand data requirements; translated business processes into efficient, scalable applications; maintained high standards in **clean, readable code** with comprehensive documentation for team knowledge transfer.

SKILLS

Web & Back-End Development – Java Spring Boot, PHP, REST APIs, Flask, FastAPI.

Database & Data Management – PostgreSQL, MySQL, Oracle, SQL, data modelling, data warehousing, ETL.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, cloud computing infrastructure.

Data Analysis & Business Intelligence – Power BI, Tableau, DAX, Power Query, R / RStudio, pandas, NumPy, scikit-learn, statistical analysis, bias & precision controls.

DevOps & Deployment – Jenkins CI/CD, Git, GitHub Actions, document release deployment, change management.

Domain Expertise – Laboratory Information Systems, data quality assurance, automated quality control, data computations, data transfer optimization.

Professional Standards – Problem-solving, attention to detail, time management, priority management, independent & team-based work, professional communication.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

PROJECTS

Laboratory Information Systems Platform Evolution (HCL Technologies)

- Architected **data quarantine systems** and **automated quality control verifications** using **Java Spring Boot** microservices and **Python automation scripts** to validate incoming instrument data and flag anomalies in real-time, reducing manual review time by 40%.
- Implemented **data computations** pipeline integrating **SQL-based ETL** across PostgreSQL and Oracle databases to standardize **statistical analysis** for **bias, precision, and uncertainty of measurement controls** across multiple laboratory sites; documented design and deployment procedures.
- Developed **Power BI dashboards** with **DAX measures** and **data modelling** to visualize instrument performance, data throughput trends, and quality metrics for cross-functional teams; optimised application for speed and scalability to handle 25% growth in daily transactions.

Multi-Site Data Integration & Cloud Migration (HCL Technologies)

- Designed and deployed **integration / data import / export systems** using **REST APIs** and **Python ETL** to synchronize laboratory data across **on premise & cloud** infrastructure using **AWS (EC2, S3, Lambda, CloudWatch)** and **Docker containerisation**.
- Established **document release deployment & change management** procedures using **Jenkins CI/CD pipelines** and **Git branching** to ensure safe rollout of data system updates across distributed laboratory sites; maintained detailed **design documents** for cross-team reference.
- Collaborated with quality assurance and field services teams to understand **data requirements** and translate them into efficient, flexible processes; wrote clear project plans and **clean, readable code** that enabled junior developers to contribute independently to ongoing maintenance.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)